



Driver Monitoring System (DMS)

Comprehensive Operational Architecture, Verification Scenarios & Safety Detection Guide

1. Executive Summary & Core Architecture

The Driver Monitoring System (DMS) is an edge AI safety application running on Flutter. It provides continuous driver verification, vigilance monitoring (drowsiness, sleep, distraction), unsafe behavior detection (phone usage, smoking, unbuckled seatbelt), and vehicle telemetry over HTTP/WebSocket cloud APIs.

Key Operational Phases:

- **Phase.verify**: Initial driver identity verification phase upon app launch or trip restart.
- **Phase.monitoring**: Active safety AI processing at ~10 FPS with real-time audio alerts and on-screen HUD banners.

2. Phase 1: Driver Verification Scenarios (Phase.verify)

Scenario 2.1: Registered Driver Verification (Success)

- **Trigger**: Enrolled driver faces front camera during verification.
- **Verification Engine**: MobileFaceNet calculates live 192-d facial embedding and measures Euclidean distance ($d < 0.90$) against database reference photos.
- **Action Taken**:
 - Voice Alert: "Welcome [Driver Name]. Driver verification successful. Please drive safely."
 - Clean 2.5-second confirmation screen displayed.
 - Session enters **Phase.monitoring** as a **Verified Driver**.

Scenario 2.2: Unregistered / Unknown Driver (Mismatch)

- **Trigger**: Unregistered person (face not in database) sits in front of camera for 10 continuous seconds.
- **Action Taken**:
 - Voice Alert: "Driver not verified. Please verify your face before driving."
 - Session enters **Phase.monitoring** as an **Unknown Driver** (`isUnknownDriver = true`).

Scenario 2.3: Speed & Accelerometer Motion Triggered Entry (> 3 km/h)

- **Trigger**: Vehicle starts moving above 3.0 km/h before facial verification completes.
- **Dual Motion Sensing**:
 - **GPS Speed Stream**: Realtime GPS speed calculation (includes distance-over-time fallback calculation).
 - **Accelerometer Sensor (sensors_plus)**: Linear 3D acceleration magnitude $a = \sqrt{a_x^2 + a_y^2 + a_z^2}$ integrated into velocity estimate (km/h). Functions inside tunnels, basements, or GPS-dead zones.

- **Action Taken:**
 - **Silent Transition:** No "Verification failed" or warning sound spoken.
 - Incident logged to cloud API: Unverified Driver.
 - Enters **Phase.monitoring** as Speed-Triggered Session.

3. Phase 2: Active Safety Monitoring Scenarios (Phase.monitoring)

Incident / Behavior	Detection Method & Threshold	Voice Alert	UI HUD Banner	API Incident Reported
Asleep / Drowsiness	Eye Aspect Ratio (EAR) drops below threshold for $\geq 1.5s$	"Wake up! Drowsiness detected"		Drowsiness
Distraction	Head Euler Yaw/Pitch $> 22^\circ$ for $> 2.0s$	"Keep your eyes on the road"		Distracted
Phone Usage	Object Detection confidence > 0.65	"Please do not use your phone..."		Phone Usage
Smoking / Cigarette	Object Detection confidence > 0.65	"No smoking while driving"		Smoking
Unbuckled Seatbelt	Buckle status false while moving	"Fasten your seatbelt"		Seatbelt
Driver Changed (Swap)	Physical face.trackingId mismatch + FaceNet $d > 0.92$ (6 frames)	"Driver changed"		Driver Changed

Scenario 3.1: Strict Physical Face Lock for Verified Driver

0% False "Driver Changed" Alerts:

Once verified, ML Kit locks onto the physical face using its hardware `face.trackingId`. As long as the verified driver remains in the seat, head turns, mirror checks, or lighting changes will **NEVER** trigger a false "Driver Changed" alert.

Driver Replacement: When a new person sits down, ML Kit assigns a new `trackingId`.

FaceNet evaluates distance $d > 0.92$ for 0.6s and fires the  **DRIVER CHANGED** alert immediately.

4. Trip Lifecycle & Closing Scenarios

Scenario 4.1: Standard Trip End (Verified / Unknown Drivers)

Closed when the camera detects **No Human (0 faces)** in view for **30 continuous seconds**. The app calls `_sendTripEnd()` API and resets to **Phase.verify**.

Scenario 4.2: Speed-Triggered Trip End (Stationary Speed ≤ 3 km/h)

Speed-Triggered Trip End:

For trips started via speed without a face frame, "No Human" alerts are suppressed. When the vehicle stops (speed ≤ 3.0 km/h / 0 km/h) for **30 continuous seconds**, the trip automatically closes, sends `_sendTripEnd()` to the cloud API, and resets to the Driver Verification screen.